and height of an object, so long as we make sufficiently accurate measurements.

Newton's *absolute time* was the time of everyday experience, the time that flows inexorably forward as we age, the time measured by high-quality clocks and by the rotation of the Earth and motion of the planets. It is a time whose flow is experienced in common by all humanity, by the Sun, by all the planets and the stars. According to Newton we all, regardless of our motion, will agree on the period of some planetary orbit or the duration of some politician's speech, so long as we all use sufficiently accurate clocks to time the orbit or speech.

If Newton's concepts of space and time as absolute were to crumble, the whole edifice of Newtonian physical laws would come tumbling down. Fortunately, year after year, decade after decade, century after century, Newton's foundational concepts had stood firm, producing one scientific triumph after another, from the domain of the planets to the domain of electricity to the domain of heat. There was no sign of any crack in the foundation—until 1881, when Albert Michelson started timing the propagation of light.

It seemed obvious, and the Newtonian laws so demanded, that if one measures the speed of light (or of anything else), the result must depend on how one is moving. If one is at rest in absolute space, then one should see the same light speed in all directions. By contrast, if one is moving through absolute space, say eastward, then one should see eastward-propagating light slowed and westward-propagating light speeded up, just as a person on an eastbound train sees eastward-flying birds slowed and westward-flying birds speeded up.

For the birds, it is the air that regulates their flight speed. Beating their wings against the air, the birds of each species move at the same maximum speed through the air regardless of their flight direction. Similarly, for light it was a substance called the *aether* that regulated the propagation speed, according to Newtonian physical laws. Beating its electric and magnetic fields against the aether, light propagates always at the same universal speed through the aether, regardless of its propagation direction. And since the. aether (according to Newtonian concepts) is at rest in absolute space, anyone at rest will measure the same light speed in all directions, while anyone in motion will measure different light speeds.